

# GoalFlow

In-House Goal Setting & Tracking Portal  
AtomQuest Hackathon 1.0 — Official Submission Document  
*"Set goals that actually get done."*

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## 1. Live / Hosted Demo URL

| Platform     | URL                                                                                                       |
|--------------|-----------------------------------------------------------------------------------------------------------|
| GitHub Pages | <a href="https://pappoppulalavanya.github.io/goalflow/">https://pappoppulalavanya.github.io/goalflow/</a> |

## 2. Source Code Repository (GitHub)

| Field          | Value                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------|
| Repository URL | <a href="https://github.com/Pappoppulalavanya/goalflow">https://github.com/Pappoppulalavanya/goalflow</a> |
| Platform       | GitHub                                                                                                    |
| Visibility     | Public                                                                                                    |
| Branch         | main                                                                                                      |
| Main App File  | <a href="#">index.html</a>                                                                                |

## 3. Architecture Diagram



Architecture Layers Covered:

| Layer             | Components                                     | Technology            |
|-------------------|------------------------------------------------|-----------------------|
| Users / Clients   | Employee · Manager · Admin                     | Browser — any device  |
| Frontend          | Login · Dashboard · Goals · Check-in · Reports | React 18 + Chart.js 4 |
| Application Logic | RBAC · Goal Engine · Validation · Workflow     | React State + Pure JS |
| Data Layer (Demo) | In-memory React state — zero backend needed    | useState              |
| Data Layer (Prod) | Relational DB with ORM                         | PostgreSQL + Prisma   |
| Auth              | Role-based access control                      | JWT + NextAuth        |
| Deployment        | Static file hosting                            | GitHub Pages          |

## 4. Login Credentials — All 3 Roles

Use the [Quick Demo Login buttons](#) on the login screen to switch roles instantly — or enter credentials below manually.

| Role       | Button Color                   | Email              | Password | Access Level       |
|------------|--------------------------------|--------------------|----------|--------------------|
| Employee   | <input type="checkbox"/> Blue  | priya@atomquest.in | emp123   | Own goals only     |
| Employee 2 | <input type="checkbox"/> Blue  | rahul@atomquest.in | emp123   | Own goals only     |
| Manager    | <input type="checkbox"/> Green | meera@atomquest.in | mgr123   | Team + Approvals   |
| Admin      | <input type="checkbox"/> Amber | admin@atomquest.in | admin123 | Full system access |

## 5 Tech Stack & Backend Approach

| Layer           | Technology             | Purpose                                |
|-----------------|------------------------|----------------------------------------|
| Frontend (Demo) | React 18 (CDN)         | Single-file SPA — zero build step      |
| Charts          | Chart.js 4.4 (CDN)     | QoQ trends, heatmap, donut charts      |
| Styling         | CSS Custom Properties  | Design tokens, dark theme, responsive  |
| Hosting         | GitHub Pages / Netlify | Static file hosting — free tier        |
| Frontend (Prod) | Next.js + TypeScript   | SSR, type safety — Vercel free tier    |
| Backend (Prod)  | Node.js + Express      | REST API, JWT auth — Railway free tier |
| Database (Prod) | PostgreSQL + Prisma    | Relational DB — Supabase 500MB free    |
| Auth            | JWT + NextAuth         | RBAC, session management, Entra SSO    |

## Backend Architecture Approach:

For the hackathon demo, the entire application runs as a **self-contained single HTML file** using React 18 loaded via CDN with in-memory state management. No backend server is required — all business logic runs client-side.

| Concern         | Approach                                                         |
|-----------------|------------------------------------------------------------------|
| API Design      | RESTful — Express.js with route-level middleware                 |
| Authentication  | JWT tokens (8h expiry) + role claim — Employee / Manager / Admin |
| Authorisation   | Middleware checks role before each route handler                 |
| Goal Validation | Server-side: max 8 goals, min 10% weight, total = 100% on submit |
| Goal Locking    | approvalStatus = 'approved' blocks PUT/DELETE at API level       |
| Check-in Logic  | Score formula applied per UoM type on the backend before storing |
| Audit Trail     | Middleware logs user, action, timestamp, target to audit table   |
| Data Export     | CSV streamed from server — Content-Disposition: attachment       |
| Scalability     | Stateless API — horizontal scale via Railway / Render / AWS ECS  |